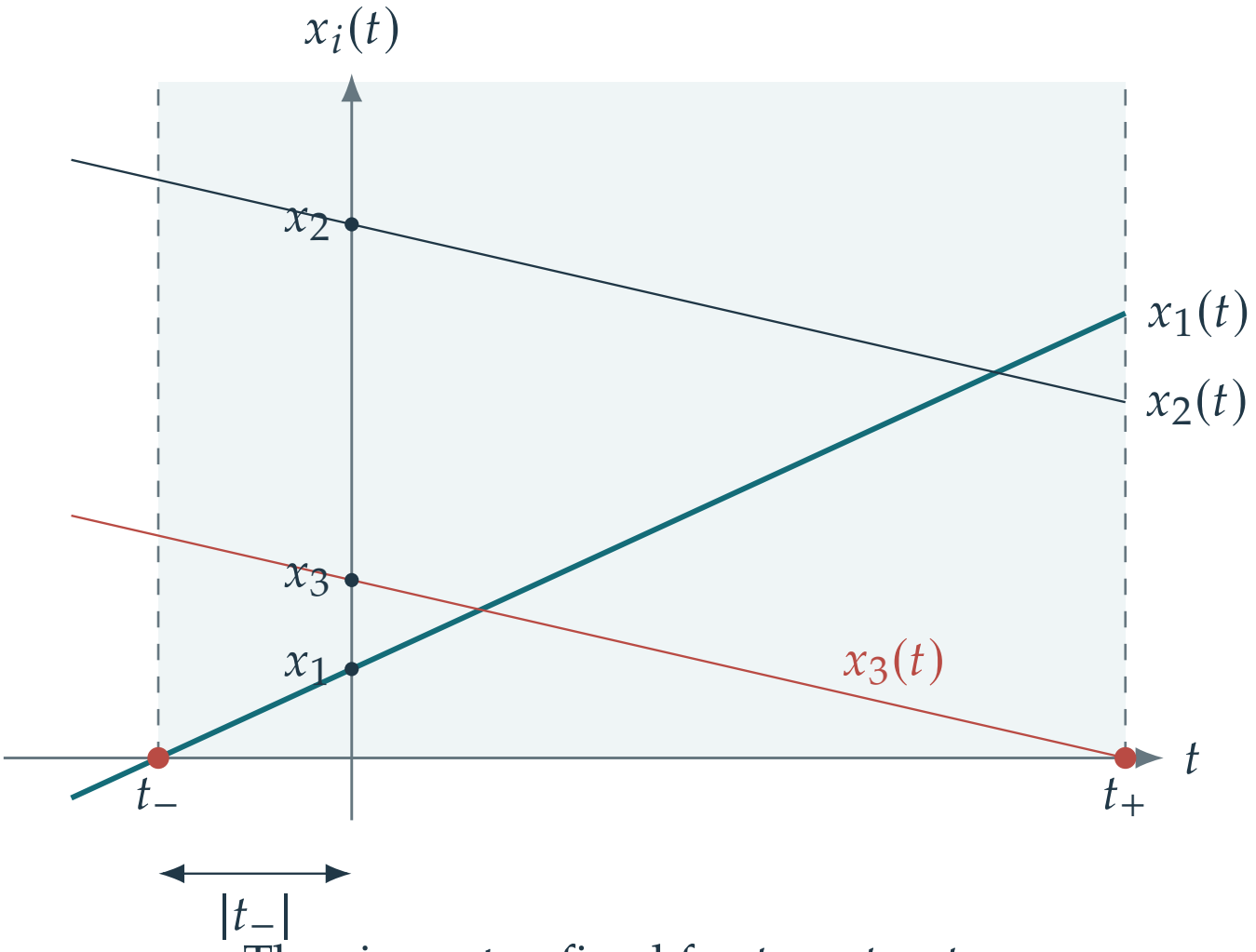

At a minimizer x:

$x(t) = x + th, Ah = 0$

$Ax(t) = Ax$

$\|x(t)\|_1 = \|x\|_1$
for $t_- \leq t \leq t_+$.

The signs stay fixed for $t_- < t < t_+$.

At either endpoint, at least one coefficient becomes zero.